

Lab 1 — Hands-On with AI Tools and Vibe Coding in R

Course: PM599: AI as a Tool for Biostatistics

Date: May 26, 2026 · 1:00–3:50 PM · SSB-114

Instructor: Lu Zhang

Before You Begin

Make sure you have all three platforms open in browser tabs:

- **ChatGPT:** chatgpt.com (sign in with your USC account)
- **Claude:** claude.ai (free account)
- **Gemini:** gemini.google.com (sign in with Google account)

And have **RStudio** (or R) open with the starter script:

```
week1/lab/starter_code/vibe_coding_demo.R
```

Lab Overview

Activity	Time	Description
Setup	1:00–1:15	Account checks, privacy settings, personalization
Activity 1	1:15–2:00	Platform comparison
Activity 2	2:00–2:45	Prompt refinement
Break	2:45–2:50	5 min
Activity 3	2:50–3:45	Vibe coding in R
Wrap-up	3:45–3:50	Debrief and HW1 overview

Setup (1:00–1:15 PM)

Privacy Settings

Follow the instructor's demonstration to configure data privacy settings:

ChatGPT:

1. Click your profile icon (top right) → Settings → Data Controls
2. Turn OFF *"Improve the model for everyone"*

Claude.ai:

- Anthropic does not use Claude.ai conversations to train models by default
- Verify by checking Settings → Privacy

Gemini:

1. Go to myactivity.google.com

Activity 1 — Platform Comparison (1:15–2:00 PM)

Objective: Compare how different LLMs respond to the same biostatistics prompts.

Format: Individual work, group discussion at the end.

Instructions

Run each of the prompts below through **at least two platforms**. Record the responses (copy-paste into a notes document or screenshot). At the end, compare with a neighbor.

Prompt 1 — Explain a concept:

Explain what a p-value means in plain language. Avoid the phrase "probability that the null hypothesis is true."

Things to look for: Is the explanation accurate? Does it avoid common misconceptions (e.g., confusing p-value with effect size)? Is it clear?

Prompt 2 — Interpret a result:

In a logistic regression model, I get an odds ratio of 2.3 with a 95% confidence interval of (1.1, 4.8) and $p = 0.03$ for the predictor "smoking status." The outcome is 5-year cardiovascular disease. Interpret this result for a clinical audience.

Things to look for: Does it correctly interpret the OR? Does it mention uncertainty (the CI is wide)? Does it note that this is an adjusted or unadjusted estimate?

Prompt 3 — Generate R code:

Write R code to simulate 200 observations from a simple linear regression model:
 $y = 2 + 0.5 * x + \text{noise}$, where $x \sim \text{Normal}(0, 1)$ and $\text{noise} \sim \text{Normal}(0, 1)$.
Then fit the model with `lm()` and print the summary.

Things to look for: Does the code run without errors? Does `set.seed()` appear for reproducibility? Are the coefficients recovered approximately correctly?

Test it: Copy the R code into RStudio and run it. Does it work?

Discussion Questions

After running the prompts, discuss with your neighbor:

1. Which platform gave the most accurate answer for Prompt 1? Did any make an error?
2. Did any platform add a disclaimer or hedge its answer? Was that appropriate?
3. For Prompt 3, did the code run correctly on the first try?
4. Would you trust these outputs in a research context without verification? Why or why not?

Activity 2 — Prompt Refinement (2:00–2:45 PM)

Objective: Improve a weak prompt step by step and observe how responses change.

Format: Pairs or individual work.

Background

A strong prompt typically provides:

- **Role:** Who should the AI act as?
- **Context:** What is the situation?
- **Task:** What exactly do you need?
- **Format:** How should the answer be structured?

Exercise

Start with this weak prompt:

Activity 3 — Vibe Coding in R (2:50–3:45 PM)

Objective: Use an AI tool to help write, explain, and extend R code for a statistical analysis — while practicing human-in-the-loop verification.

Format: Individual, with instructor demonstrations.

Setup

Open the starter script in RStudio:

```
week1/lab/starter_code/vibe_coding_demo.R
```

Run through **Sections 1–4** to understand the base analysis:

- Section 2: Simulates a clinical trial dataset (SBP before/after)
- Section 3: Produces a box plot of the outcome by group
- Section 4: Runs a two-sample t-test

Wrap-Up and HW1 (3:45–3:50 PM)

Quick debrief — show of hands:

- Who found a factual error in an AI response today?
- Who had AI-generated code that didn't run on the first try?
- Who was surprised by how well (or how poorly) the AI performed?

Homework 1: Compare AI tools on a biostatistics task of your choice.

Full instructions at homework/hw1.md.

Due: May 28, 2026 before class. Submit via Brightspace.

Key Takeaways from Today

1. Different LLMs perform differently on the same prompt — always verify.
2. Prompt quality matters enormously; small changes can dramatically improve responses.
3. AI-generated code runs $\neq$ AI-generated code is correct.
4. Privacy settings are a non-negotiable first step before any research use.
5. You — not the AI — are responsible for the accuracy of your analysis.